Narek Harutyunyan

narekharutyu.github.io | Google Scholar | narek_harutyunyan@berkeley.edu

Education

University of California, Berkeley **Aug. 2026–May 2031 (expected)**
Ph.D. in Electrical Engineering and Computer Sciences *Berkeley, CA*

Brown University **Sept. 2022–May 2026**
B.S. in Computer Engineering, ***GPA: 4.0/4.0*** *Providence, RI*

Research Experience

Robotics Institute Summer Scholar **May 2026–Present**
Carnegie Mellon University (CMU) — LeCAR Lab, Prof. Guanya Shi *Pittsburgh, PA*

- Developing a cross-embodiment visuomotor diffusion policy from wrist-camera demonstrations, using a shared task-space action representation across wheeled, legged, and aerial manipulators.
- Built six cross-embodiment mobile-manipulation tasks with matched Isaac Lab and hardware setups, plus pipelines for multimodal demonstration collection, policy training, and evaluation.
- Developed IK-, MPC-, and RL-based whole-body controllers to track policy-generated task-space trajectories on wheeled, legged, and aerial manipulators.

Summer Undergraduate Research Fellow **May–Sept. 2025**
California Institute of Technology (Caltech) — ARCL, Prof. Soon-Jo Chung *Pasadena, CA*

- Developed **ContractionPPO**, an RL framework with differentiable contraction layers that provides certified stability for learned whole-body controllers.
- Built a 6-DoF moving-platform environment in Isaac Lab and a complementary 3-DoF physical testbed to evaluate policies under controlled platform disturbances.
- Built a sim-to-real SDK for closed-loop deployment of Isaac Lab policies on Unitree Go1 and Booster T1.
- Deployed ContractionPPO on the Unitree Go1 and Booster T1 for stabilization, locomotion, acrobatics, and fault recovery under platform motion and wind — including field tests aboard an inflatable boat.
- Achieved 100% task success across 2,500 simulation and 10 hardware trials for quadruped handstand stabilization on moving platforms, withstanding wind disturbances up to 9.6 m/s.

Robotics Institute Summer Scholar **May 2024–July 2025**
Carnegie Mellon University — AirLab, Prof. Sebastian Scherer *Pittsburgh, PA*

- Refactored **MapEx**, a probabilistic frontier-based exploration algorithm, into a Gym-compatible environment and eliminated agent-backtracking failures.
- Proposed and led **MapExRL**, an RL framework combining frontier-based planning, predicted global maps, and uncertainty modeling for long-horizon exploration.
- Built an IoU-based evaluation suite showing that MapExRL reached equivalent mapping accuracy with up to 18.8% fewer exploration steps than prior baselines on real-world indoor maps.

Undergraduate Researcher **May 2023–May 2026**
Brown University — Automatic Coordination of Teams Lab, Prof. Nora Ayanian *Providence, RI*

- Trained multi-agent RL policies in Isaac Lab for cooperative object transport and obstacle avoidance with Crazyflie quadrotor swarms.
- Quantified downwash-induced forces, moments, and velocities during proximity flight using Particle Image Velocimetry and six-axis force/torque sensing; published at AIAA Aviation Forum and ASCEND.
- Engineered an aerial-painting quadrotor and control API deployed as a recurring course platform for three years, supporting hands-on instruction for 75+ students.
- Managed and maintained a research fleet of 50+ quadrotors, quadrupeds, and mobile robots for laboratory experiments and teaching.

Selected Publications

- **ContractionPPO: Certified Reinforcement Learning via Differentiable Contraction Layers.** V. Zinage*, **N. Harutyunyan***, E. Verheyden*, et al. *IEEE Robotics and Automation Letters (RA-L)*, 2026. [Paper] [Project] [Video]
- **MapExRL: Human-Inspired Indoor Exploration with Predicted Environment Context and Reinforcement Learning. N. Harutyunyan***, B. Moon*, S. Kim, et al. *IEEE International Conference on Advanced Robotics (ICAR)*, 2025; also presented at the ICRA 2025 Workshop. [Paper] [Project] [Video]
- **MapEx: Indoor Structure Exploration with Probabilistic Information Gain from Global Map Predictions.** C. Ho*, S. Kim*, B. Moon, A. Parandekar, **N. Harutyunyan**, et al. *IEEE International Conference on Robotics and Automation (ICRA)*, 2025. [Paper] [Project] [Video]
- **Downwash Dynamics: Impact of Quadrotor Separation on Forces, Moments, and Velocities for Dense Formation Flight.** A. Kiran, **N. Harutyunyan**, N. Ayanian, K. Breuer. *AIAA Aviation Forum and ASCEND*, 2024. [Paper]

** denotes equal contribution.*

Technical Skills

Programming & ML: Python, C++; PyTorch, TensorFlow; Vision Transformers/CLIP, U-Net Diffusion Models; GPU-Accelerated Multi-GPU Training
Robot Learning: Reinforcement Learning, Imitation Learning, Diffusion Policies, Visuomotor Learning, Sim-to-Real Transfer, Cross-Embodiment Transfer
Control & Planning: Whole-Body Control, Stability-Certified Control, IK, MPC
Robotics & State Estimation: ROS/ROS2, Isaac Lab/Isaac Sim, MuJoCo; Visual-Inertial Odometry (VIO), LiDAR-Aided VIO, SLAM
Hardware & Prototyping: Ridgeback–Franka, Unitree Go1/Go2, Booster T1, YAM Bimanual Arms; Motion Capture (Vicon, OptiTrack), Force/Torque Sensing; CAD (Fusion 360, AutoCAD), 3D Printing

Teaching & Leadership

Head TA, Creating Art with Teams of Robots — Brown University Spring/Fall 2024; Spring 2026

- Led weekly laboratories and mentored final projects for 75+ students using quadrotors, quadrupeds, and ground robots.

Teaching Assistant, Deep Learning — Brown University Fall 2025

- Designed PyTorch and TensorFlow labs and assignments, taught workshops on transformers, representation learning, diffusion models, RL, and robotics, and mentored final projects for 120+ students.

Teaching Assistant, Introduction to Engineering — Brown University Fall 2023

- Led technical workshops for 100+ students on MATLAB, Fusion 360, rapid prototyping, machine-shop tools, and Arduino electronics; supported 40+ students in completing multidisciplinary final projects.

Honors & Professional Service

Honors: Berkeley Fellowship for Graduate Study (2026; 1 of 3 EECS recipients); CMU Robotics Institute Summer Scholar (2024, 2026); Caltech Summer Undergraduate Research Fellowship (2025); Brown Undergraduate Teaching and Research Awards (2023, 2024); Tau Beta Pi Engineering Honor Society (2024–Present); Huys Foundation STEM Scholar (2023, 2024; $\approx$ 5% selection rate).
Peer Reviewer: IEEE Robotics and Automation Letters (RA-L), 2026; International Conference on Advanced Robotics (ICAR), 2025.